Fairness-Aware Online Learning Under Concept Drift With an Adaptive Fairness Constraint

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***Abstract*—Online learning systems in nonstationary environments must stay both predictive and fair as the data distribution evolves, yet existing fairness-aware learners optimize a fixed regularizer and ignore concept drift, while drift-adaptive streaming methods enforce no fairness. We propose Fairness-Aware Drift-Aware Online Learning (FADO), which couples the Aranyani [1] fair oblique decision forest with an adaptive fairness constraint and a drift reaction controller. Instead of weighting the fairness penalty by a constant chosen in advance, FADO bounds demographic parity and treats the penalty weight as the Lagrange multiplier of that bound tracked, online, by stochastic dual ascent on a per-sample Horvitz–Thompson estimate of each group's disparity that we prove to be unbiased. A dual-threshold ADWIN detector and a fairness monitor reset the penalty's running statistics on confirmed drift, and the reaction controller spikes the learning rate and sharpens the forest's routing through a temperature that FADO introduces, annealing both back during recovery.**

***Index Terms*—fairness-aware online learning, drift, constrained learning, drift detection, group fairness, oblique decision forests**

I. Introduction

The increasing deployment of machine learning (ML) systems in high-stakes decision-making domains, including healthcare [2], criminal justice [3], and hiring [4], where their predictions may directly affect individuals and society, makes optimizing for performance an insufficient objective; models must also satisfy fairness requirements to ensure equitable treatment across individuals or groups of individuals [5]. In this regard, among the fairness literature, *group fairness* [6] has emerged as one of the most widely studied paradigms, aiming to ensure that predictive behavior is balanced across populations defined by sensitive attributes such as gender, race, or age. Common formulations include demographic parity (DP), equal opportunity, and equalized odds (EO), each imposing statistical constraints on model outcomes across groups (sometimes mutually incompatible) [6], [7]. Most existing fairness-aware models, however, are developed under the assumption of a stationary data distribution. In practice, real-world environments are inherently dynamic, and the statistical properties of data may evolve over time due to policy changes, behavioral shifts, seasonal effects, economic fluctuations, or other social transformations. This phenomenon, commonly referred to as concept drift, occurs when the relationship between input variables and target outcomes changes over time, causing previously learned decision boundaries to become outdated. Different forms of concept drift [8] require different adaptation strategies and can degrade performance when models fail to adjust. Drift also threatens fairness: shifting distributions alter the statistical relationships that support fairness guarantees, increasing bias against some groups. To address this challenge, the online learning and data stream mining literature has proposed numerous adaptive methods capable of detecting and responding to concept drift in real time. These approaches often rely on drift detection mechanisms combined with incremental learning strategies to maintain predictive accuracy under nonstationary conditions. Nevertheless, the majority of existing methods focus primarily on predictive utility [9], [10], while fairness considerations remain comparatively underexplored. As a result, fairness guarantees established during training may deteriorate over time even when predictive accuracy remains relatively stable [11]. This limitation is particularly critical in high-stakes applications, where persistent fairness violations may remain undetected for extended periods, compromising trustworthiness, reinforcing societal inequalities, and causing tangible harm to affected groups. These limitations highlight an important gap in the literature and motivate the following research questions (RQs):

- **RQ1:** To what extent does concept drift affect group fairness metrics in online learning settings?
- **RQ2:** Can adaptive drift detection and response mechanisms preserve fairness guarantees while maintaining predictive performance over time?

Based on these research questions, this work investigates fairness-aware online learning under concept drift. We introduce **F**airness-**A**ware **D**rift-Aware **O**nline Learning (FADO), a framework that leverages an existing soft-routed fair oblique decision forest, Aranyani [1], extending it with three mechanisms: an *adaptive fairness constraint*, which replaces the fixed fairness weight of the base learner with the Lagrange multiplier of a bound on the fairness metric and tracks it online; a dual-threshold drift detector, complemented by a fairness monitor, whose confirmed drifts reset the fairness penalty's running statistics; and a reaction controller that modulates the optimizer learning rate and a routing temperature, which FADO adds to the forest, in response to detected change. We emphasize that ARANYANI is not contributed by this paper: it serves as the in-processing base learner around which FADO builds its constraint, drift-detection, and response machinery. Our central hypothesis is that fairness degradation under concept drift can be mitigated by enforcing fairness as a constraint whose price adapts to the data, coupled with explicit drift detection and response, allowing the model

to maintain both predictive utility and equitable treatment across groups in nonstationary environments.

To evaluate this hypothesis, we run our evaluation on two complementary testbeds. FOLKTABLES (American Community Survey (ACS) Income, California) contributes a natural cross-year shift between 2015 (training) and 2017–2018 (prequential test), and ADULT serves as a controlled testbed into whose stream we inject, the four main types of concept drift we can find in the real world [12] (although many others exist). FADO is compared against three streaming baselines spanning the accuracy/fairness/adaptivity design space: the Adaptive Random Forest (ARF) [9], an accuracy-oriented ensemble with drift detection but no fairness mechanism; robust fairness regularization (RFR) [13], a sharpness-aware fairness-reweighted method without adaptive drift handling; and a controller-free ARANYANI baseline, identical to the base learner inside FADO but with a fixed fairness weight and no drift handling, which isolates FADO's contribution. The main contributions of this work are:

- FADO, a fairness-aware online learning framework that wraps a fair oblique decision forest with an adaptive fairness constraint, a dual-threshold ADWIN drift detector with a fairness monitor, and a reaction controller acting on the learning rate and on a routing temperature that FADO introduces;
- Where ARANYANI relaxes its node-level fairness constraints into a penalty with a fixed weight, a formulation that keeps a bound on the demographic parity of the predictions and treats the penalty weight as its dual price, tracked online, together with a per-sample Horvitz–Thompson estimate of each group's disparity that we prove unbiased and that drives both the price and the fairness monitor;
- An empirical analysis over 30 seeds, on a natural cross-year shift on FOLKTABLES and on four injected drift types on ADULT, of how concept drift affects group fairness over the stream, comparing FADO against the ARF, RFR, and controller-free ARANYANI baselines with paired, multiplicity-corrected tests, and analyzing FADO's runtime overhead.

II. Related Work

Following the most adopted taxonomy [14], in this section, the works will be divided into three categories: (i) pre-processing methods, (ii) in-processing methods, and (iii) post-processing methods.

A. Pre-Processing Methods

Pre-processing methods aim to mitigate bias in the data before the training stage of a model. Xiong et al. [15] propose leveraging training samples reweighting to satisfy demographic parity while minimizing distributional distortion using the Wasserstein distance to the original dataset. Although effective, this method does not account for concept drift, nor does it prove robustness to other forms of fairness violations, which limits its applicability in dynamic environments. Zeng et al. [16] propose a different approach that contrasts with the aforementioned, modifying the actual data labels by treating the removal of discrimination from the training dataset as a randomized response problem. The main contribution of this work is the provision of a theoretical bridge between pre-processing techniques and downstream Bayes-optimal classifiers, and the fact that it is model agnostic: because it only alters the training labels, users do not need to modify their current pipelines. However, this method also does not account for concept drift or its impact, and it is limited by the binary setup, meaning it only works with binary sensitive attributes and binary outcomes.

B. In-Processing Methods

In-processing methods incorporate fairness constraints directly into the learning algorithm. Four lines are especially relevant to our work.

Chowdhury et al. [1] propose ARANYANI, an online fair-learning ensemble of soft-routed oblique decision trees in which the demographic-parity penalty is enforced at the level of individual routing nodes, inducing a parameter-isolation property that keeps the per-step fairness gradient cheap and lets the ensemble track running per-group statistics rather than storing past samples. ARANYANI starts from node-level fairness constraints but relaxes them into a regularizer whose weight is a fixed hyperparameter, and it has no mechanism for drift. We adopt it as our base learner; these two limitations motivate our extension.

Gomes et al. [9] propose ARF, a streaming ensemble of Hoeffding trees with per-tree warning and drift detectors that swap in background trees on confirmed drift. ARF is the standard streaming-adaptation baseline but is purely accuracy-oriented; we treat it as a representative adaptive-but-not-fair baseline.

Jiang et al. [13] model distribution shift as a bounded weight-space perturbation and optimize for worst-case fairness under that perturbation, producing the sharpness-aware reweighting (RFR) that we use as our fair-but-not-adaptive baseline.

Finally, fairness can be imposed as a constraint rather than as a weighted penalty. Zafar et al. [17] bound a covariance proxy of disparate impact, Agarwal et al. [18] reduce fair classification to a saddle-point problem solved by a sequence of cost-sensitive learners, and Cotter et al. [19] optimize non-differentiable rate constraints with a proxy-Lagrangian game. These methods solve the problem offline on a fixed training set; online convex optimization with long-term constraints [20] provides the primal–dual template that FADO follows one sample at a time.

C. Post-Processing Methods

Post-processing approaches transform an already-trained classifier's outputs. Hardt et al. [7] introduce randomized threshold adjustments that equalize group error rates and remain the standard model-agnostic baseline; Mishler et al. [21] extend the idea to counterfactual equalized odds in intervention

settings (at the cost of strong causal assumptions); Small et al. [22] replace fixed randomization regions with smooth monotonic probability curves to recover individual fairness. None of these methods touch the training loop, so none of them adapt under drift; we mention them for completeness rather than as direct comparators.

D. Main Limitations of Prior Work

Notwithstanding the significant contributions of the families surveyed above, we identify four main limitations that jointly motivate this work and the design of FADO:

- **L1 – Static fairness assumption.** Once the protected-attribute conditional or the label distribution shifts, the fairness guarantees obtained at training time provide no protection at deployment;
- **L2 – Adaptive methods are accuracy-oriented.** Streaming ensembles such as ARF [9] explicitly detect and react to drift but do so purely on the predictive-error signal, with no mechanism for tracking or correcting fairness violations;
- **L3 – Existing fair online learners do not react to drift.** ARANYANI [1] provides memory-efficient online fairness, but its regularizer is nonadaptive: when the distribution moves it continues to optimize the same penalty with the same fixed weight, although the weight that keeps disparity acceptable changes with the data, and convergence to the new optimum is slow;
- **L4 – Drift-aware fair methods rely on assumptions that do not transfer to streams.** Approaches such as [13] treat distribution shift as a bounded weight-space perturbation, which is useful in worst-case analysis but does not yield an actionable online detector or controller, and constrained fair learners [17]–[19] solve their problem offline.

FADO addresses L1–L4 directly by retaining the node-level fairness regularizer of ARANYANI as the in-processing base learner and adding on top of it (i) an adaptive fairness constraint, whose dual price replaces the fixed fairness weight; (ii) an ADWIN-based dual-threshold detector on the online error stream and a fairness monitor on a per-sample disparity signal, whose confirmed drifts reset the regularizer's running statistics; and (iii) a reaction controller that modulates the optimizer learning rate and a routing temperature added to the forest on warning and confirmation events.

III. Problem Statement

A. Online Learning Setting

We consider an online binary classification problem over a data stream $\{(x_t, y_t, a_t)\}_{t=1}^{T}$, where $x_t \in \mathbb{R}^d$ is the feature vector observed at time t, $y_t \in \{0, 1\}$ is the binary target label, and $a_t \in \mathcal{A}$ is a sensitive attribute taking values in a finite set with $|\mathcal{A}| = K \geq 2$. Following the formulation of previous work [1], [23], at each time step the learner receives x_t, produces a prediction $\hat{y}_t = \arg\max f_t(x_t)$, then observes (y_t, a_t) and updates its parameters under a prequential, test-then-train protocol. The fairness formulation introduced below supports arbitrary K; in our study each dataset has one binary protected attribute (sex), so $K = 2$.

B. Group Fairness Metrics

We focus on *group fairness*, which assesses whether model behavior is statistically similar across protected groups. ARANYANI [1] defines demographic parity on the forest's outputs for two groups and extends its formulation to more groups and to equalized odds in its appendix; we measure fairness on the *predictions*, in a form that holds for any number of groups. Let $\mathcal{A}_t \subseteq \mathcal{A}$ denote the set of sensitive groups observed within the current evaluation window, and let $\hat{p}_a = \mathbb{E}[\hat{y} \mid a]$ be the positive-prediction rate for group $a \in \mathcal{A}_t$. We define the unweighted mean group rate as

$$\bar{p} = \frac{1}{|\mathcal{A}_t|} \sum_{a' \in \mathcal{A}_t} \hat{p}_{a'}, \tag{1}$$

i.e., the average positive-prediction rate across the observed groups. The *demographic parity violation* is then the worst-case deviation of any group's positive-prediction rate from this mean,

$$D_a = \bar{p} - \hat{p}_a, \qquad \mathrm{DP} = \max_{a \in \mathcal{A}_t} |D_a|, \tag{2}$$

which is bounded between 0 (perfect parity) and 1. DP measures bias in the prediction marginal and is sensitive to differences in base rates across groups. We adopt this mean-from-average form rather than the more common pairwise gap $\max_{a,a'} |\hat{p}_a - \hat{p}_{a'}|$ because it mirrors the node-level regularizer as we implement it (Section IV-A). For a binary attribute ($K = 2$) this form reduces to $\mathrm{DP} = \frac{1}{2} |\hat{p}_0 - \hat{p}_1|$, equivalent up to a factor of two to the textbook pairwise gap.

We also report *equalized odds*, the same deviation measured within each true-label class. With $\hat{p}_{a|y} = \mathbb{E}[\hat{y} \mid a, y]$ and $\bar{p}_y$ its mean across the observed groups,

$$\mathrm{EO} = \max_{y \in \{0,1\}} \max_{a \in \mathcal{A}_t} \left| \hat{p}_{a|y} - \bar{p}_y \right|. \tag{3}$$

Restricting it to $y{=}1$ gives equal opportunity. EO separates bias from differences in base rates. We report it but do not optimize it: ARANYANI derives the label-conditional node regularizer [1, App. C.1], and constraining EO in FADO would also need a label-conditional version of the per-sample signal (10).

C. From a Fixed Penalty to a Fairness Constraint

ARANYANI [1] states fairness as constraints on the forest's internal routing nodes, but then relaxes them into a regularized objective $\mathcal{L}_{\mathrm{CE}} + \lambda \mathcal{L}_{\text{fair}}$ (Section IV-A) whose weight λ is fixed in advance, a knob with no meaning outside the dataset on which it was tuned. What a deployment can state and defend instead is a tolerance: the model's demographic parity must not exceed ε. We therefore pose

$$\min_{\theta} \ \mathbb{E}\big[\mathcal{L}_{\mathrm{CE}}(\theta)\big] \quad \text{s.t.} \quad \mathrm{DP}(\theta) \leq \varepsilon, \tag{4}$$

and ask the learner to satisfy it *at every point of a drifting stream*. Under drift the multiplier that makes (4) bind changes

over time: the weight that keeps disparity acceptable before a shift is too small after a shift that widens the gap between groups, and needlessly costly when the stream is already fair, so it must be estimated online rather than tuned once.

IV. Methodology and Experimental Design

FADO builds on the ARANYANI fair oblique forest [1], which we use as the base learner and summarize only as far as FADO relies on it (Section IV-A). On top of it, FADO adds three mechanisms (Fig. 1): (i) an adaptive fairness constraint, whose dual price replaces the base learner's fixed fairness weight (Section IV-B); (ii) drift detection with forgetting, in which a dual-threshold ADWIN detector on the error stream and a fairness monitor on a per-sample disparity signal reset the fairness statistics on confirmed drift; and (iii) a reaction controller on the learning rate and on a routing temperature that FADO adds to the forest (both in Section IV-C). Sections IV-D–IV-I then describe datasets, drift scenarios, baselines, hyperparameter selection, the evaluation protocol, and reproducibility.

Fig. 2 states the resulting prequential loop in full; the subsections that follow detail each component.

All experiments were run in Python 3.11 across several desktop machines **[TODO: list machines]**. Per-seed runs were dispatched across the machines for wall-clock parallelism; within any given seed (for a certain scenario), all four models were executed on the same machine to preserve the seed-pinned data ordering required by the paired statistical tests (Section IV-H). Source code, configuration files, and seed manifests are released for reproducibility (Section IV-I).

A. Base Learner

We use ARANYANI [1] as the base learner and refer to [1] for its details. It is an ensemble of M soft oblique trees of depth h whose internal node j routes an input to its left child with probability $n_j(x) = \sigma(w_j^\top x + b_j)$ and to its right child otherwise. Fairness is imposed on these routing decisions: ARANYANI constrains the deviation $F_{j,a}$ of each node's average routing decision for group a from its average over all inputs (for two groups, the difference between the groups' averages), the per-node analogue of D_a in (2), and then relaxes the constraints into the regularized objective

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda\, \mathcal{L}_{\text{fair}}, \qquad \mathcal{L}_{\text{fair}} = \sum_j \frac{1}{K} \sum_{a \in \mathcal{A}} \phi_\delta\big(F_{j,a}\big), \tag{5}$$

where $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss, ϕ_δ is a Huber surrogate, and λ is a fixed hyperparameter. It estimates the fairness gradient online from running per-group averages of the routing decisions and of their gradients, the statistics S, so no past sample is stored. Our implementation, used identically by FADO and the ARANYANI baseline, takes the cross-group mean unweighted and gives the linear branch of ϕ_δ a unit slope.

FADO changes this learner in three places. First, the fixed weight λ becomes the price λ_t of Section IV-B. Second, the statistics S are reset when a drift is confirmed. Third, FADO adds a routing temperature, $n_j(x) = \sigma\big((w_j^\top x + b_j)/\tau\big)$, which recovers ARANYANI at $\tau = 1$ and which the reaction controller lowers after a confirmed drift to make routing nearly hard (Section IV-C).

B. Adaptive Fairness Constraint

FADO solves (4) online by treating the weight λ in (5) as a Lagrange multiplier.

One-Sided Linear Constraints: Since $|x| \le \varepsilon$ holds exactly when $x \le \varepsilon$ and $-x \le \varepsilon$, the constraint DP $\le \varepsilon$ is equivalent to the $2K$ one-sided constraints

$$+D_a \le \varepsilon, \qquad -D_a \le \varepsilon, \qquad a \in \mathcal{A}. \tag{6}$$

The rewrite has no absolute value and no maximum, and it states *which side* of the mean a group falls on. Its purpose is statistical: each constraint is linear in D_a, so an unbiased estimate of D_a is an unbiased estimate of every constraint's value. An absolute value would not preserve this — $\mathbb{E}|v| \neq |\mathbb{E}v|$ — and a controller fed with the magnitude of a noisy disparity estimate would read noise as unfairness and raise the price indefinitely on a perfectly fair model.

Lagrangian and Dual Ascent: Attaching a multiplier $\mu_{a,\pm} \ge 0$ to each constraint gives the Lagrangian

$$\mathcal{L}(\theta, \mu) = \mathcal{L}_{\text{CE}}(\theta) + \sum_a \Big[\mu_{a,+}(D_a - \varepsilon) + \mu_{a,-}(-D_a - \varepsilon)\Big], \tag{7}$$

and (4) is the saddle point $\min_\theta \max_{\mu \ge 0} \mathcal{L}$ [24]. The gradient of $\mathcal{L}$ in $\mu_{a,\pm}$ is the violation $\pm D_a - \varepsilon$, so projected ascent reads

$$\mu_{a,\pm} \leftarrow \Pi_{[0,\lambda_{\max}]}\big(\mu_{a,\pm} + \eta_\lambda(\pm D_a - \varepsilon)\big), \tag{8}$$

interleaved with the usual gradient step in θ (primal–dual descent–ascent [20]). In control terms, μ *integrates* the violation: it keeps rising while a group's deviation exceeds ε and stops only when it does not, so it has no steady-state error. When a group's fairness is not at the limit, its multiplier is zero so if the system is consistently fair, penalty weights will gradually decrease. The projection keeps the prices nonnegative, as duality requires, and bounded to protect accuracy. FADO sums the prices into a single weight for ARANYANI's node-level penalty:

$$\lambda_t = \min\Big(\lambda_{\text{base}} + \sum_{a,\pm} \mu_{a,\pm},\ \lambda_{\max}\Big), \tag{9}$$

where λ_{base} is the fixed weight used by the ARANYANI baseline. FADO therefore never applies less fairness pressure than ARANYANI, and adds pressure only while the constraint is violated. The floor is not needed by the method: with $\lambda_{\text{base}} = 0$, the weight is set entirely by the prices, that is, by the constraint. We use $\lambda_{\text{base}} > 0$ only to make the comparison with ARANYANI fair, since both models then start from the same weight and FADO's gain cannot come from a larger initial penalty. Penalizing the nodes is a sound way to reduce output disparity: ARANYANI shows that when every node treats the groups alike, the demographic parity of the forest's output is bounded [1, Lemma 1].

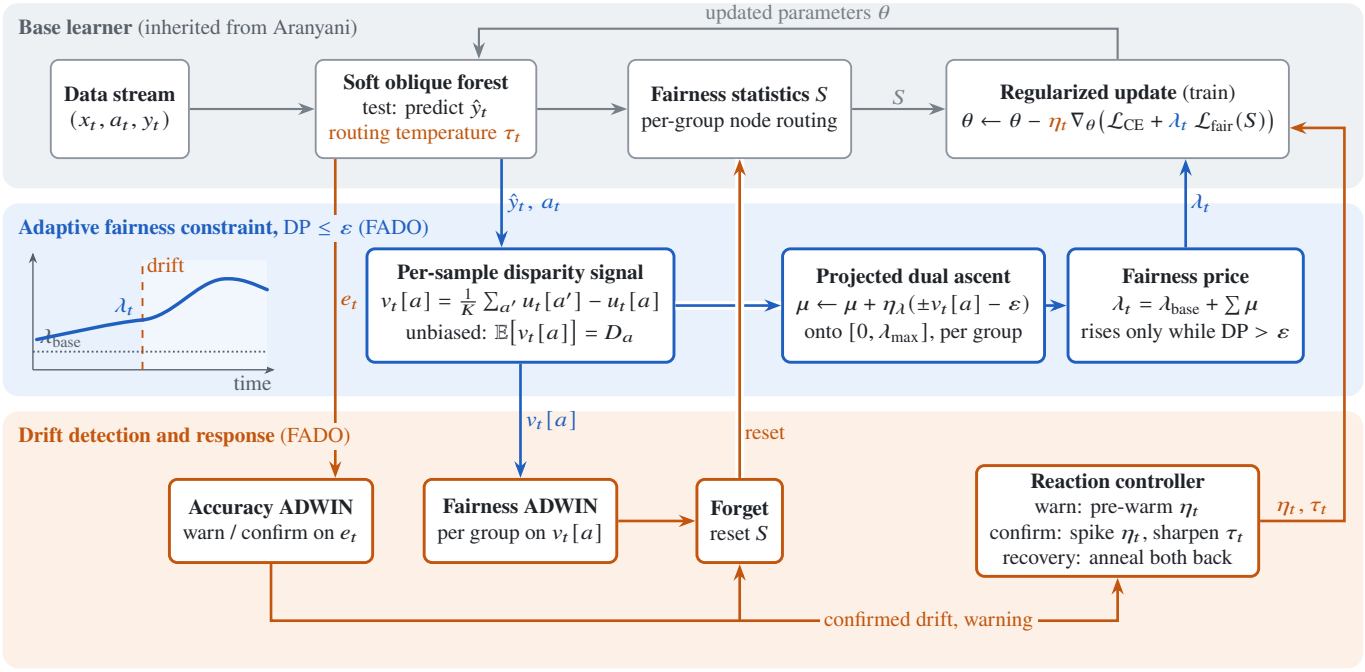


Fig. 1. Overview of FADO. Gray: the ARANYANI base learner [1], which predicts and then trains on each sample. Blue: the adaptive fairness constraint, which turns each prediction into a disparity signal and sets the fairness price λ_t. Orange: drift detection and response, which resets the fairness statistics S on drift and adjusts the learning rate η_t and routing temperature τ_t. Colored symbols in gray boxes are set by FADO.

Require: stream $\{(x_t, y_t, a_t)\}$; settings of Table II
init forest f, optimizer at η_0, temperature τ, multipliers $\mu \leftarrow 0$, running fairness stats S
for $t = 1$ **to** T **do**
  $\hat{y}_t \leftarrow \arg\max f(x_t)$; $e_t \leftarrow \mathbf{1}[\hat{y}_t \neq y_t]$ ▷ test
  update windowed accuracy, DP, and EO over the last W samples
  noise $\leftarrow e_t \wedge (\text{conf}(\hat{y}_t) > \nu)$ ▷ label-noise guard
  ok $\leftarrow \neg$*noise* $\wedge\ (t - t_{\text{last}} \geq C)$ ▷ guard, cooldown
  if $D_w(e_t)$ warns **and** *ok* **then** $\eta \leftarrow \beta_w\, \eta_0$ ▷ pre-warm
  end if
  if $D_c(e_t)$ confirms **and** *ok* **then**
    $\eta \leftarrow \beta_c\, \eta_0$; $\tau \leftarrow \tau_{\text{drift}}$; reset S; $t_{\text{last}} \leftarrow t$ ▷ spike + sharpen
  end if
  compute $v_t[a]$ by (10); reset S if a fairness detector confirms
  $\mu_{a,\pm} \leftarrow \Pi_{[0,\lambda_{\max}]}\big(\mu_{a,\pm} + \eta_\lambda(\pm v_t[a] - \varepsilon)\big)$ ▷ dual ascent
  $\lambda_t \leftarrow \min(\lambda_{\text{base}} + \sum \mu,\ \lambda_{\max})$
  if recovered **or** timed out **then** anneal $\eta \to \eta_0$ over Δ, $\tau \to 1$
  accumulate (x_t, y_t, a_t) into S ▷ train
  optimizer step on $\nabla\big(\mathcal{L}_{\text{CE}}(f(x_t), y_t) + \lambda_t\, \mathcal{L}_{\text{fair}}(S)\big)$ at rate η
end for

Fig. 2. The FADO prequential test-then-train loop.

A Per-Sample Estimate of the Violation: Online, D_a is never observed: each step reveals one individual, from one group. We therefore build, from that single sample, a quantity that equals D_a in expectation. Let $\pi_a = \Pr(a_t = a)$ and define

$$u_t[a] = \frac{\hat{y}_t\, \mathbf{1}[a_t = a]}{\pi_a}, \qquad v_t[a] = \frac{1}{K}\sum_{a'} u_t[a'] - u_t[a]. \quad (10)$$

Proposition 1 (Unbiased per-sample disparity): Conditional on the current model, if (x_t, a_t) is drawn from the stream distribution, then $\mathbb{E}[u_t[a]] = \hat{p}_a$ and $\mathbb{E}[v_t[a]] = D_a$ for every group a. Consequently $\pm v_t[a] - \varepsilon$ is an unbiased estimate of the violation of each constraint in (6).

Proof: $u_t[a]$ is nonzero only if $a_t = a$ and $\hat{y}_t = 1$, an event of probability $\pi_a \hat{p}_a$, in which case it equals $1/\pi_a$. Hence $\mathbb{E}[u_t[a]] = \hat{p}_a$, the Horvitz–Thompson identity [25]. By linearity, $\mathbb{E}[v_t[a]] = \frac{1}{K}\sum_{a'} \hat{p}_{a'} - \hat{p}_a = D_a$. The constraints in (6) are linear in D_a, which gives the last claim. ■

Substituting v_t for D_a in (8) yields a stochastic dual ascent whose steps are correct on average. The signal needs no window, so it responds from the first sample after a change, and successive values are conditionally independent, whereas a rolling disparity over W samples is almost perfectly autocorrelated. In practice π_a is replaced by the running share $\hat{\pi}_a$ (floored at 0.02), which makes (10) a consistent ratio estimator.

Remark 1 (Expected dynamics): Under a sustained violation $\Delta_{\text{DP}} = \text{DP} - \varepsilon > 0$, the active multipliers grow by about $\eta_\lambda \Delta_{\text{DP}}$ per sample; once DP $< \varepsilon$ they drain by at most $\eta_\lambda \varepsilon$ per sample, so draining is slow by design. With two groups, $D_0 = -D_1$, so the constraints in (6) coincide in pairs; our implementation keeps all four multipliers, which leaves the duality intact but makes the effective dual step $2\eta_\lambda$.

C. Drift Detection and Reaction Controller

The controller observes the online error stream $e_t = \mathbf{1}[\hat{y}_t \neq y_t]$ through two independent adaptive windowing (ADWIN) detectors [26]: a pre-warm detector D_w and a confirmation detector D_c, parameterized by significance levels δ_w and $\delta_c < \delta_w$ (in ADWIN a larger δ is more sensitive, so the pre-warm detector fires first). Both are paired with a label-noise guard that excludes a sample from triggering an event whenever an

incorrect prediction is emitted with post-softmax confidence above ν, preventing single confident-but-wrong samples from spuriously activating the controller. A minimum of m_0 samples per detector is enforced before the first activation to avoid early-stream false positives.

When D_w fires, the optimizer learning rate is multiplied by β_w, pre-warming the parameters for a potential change. When D_c subsequently confirms a drift, four actions are taken atomically: (i) the learning rate is set to $\beta_c \cdot \eta_0$; (ii) the routing temperature that FADO adds (Section IV-A) is set in every tree to τ_{drift}, sharpening routing decisions while the ensemble refits; (iii) the running fairness statistics S of Section IV-A are reset and rebuilt from post-drift samples; and (iv) both detectors are reset and a cooldown of C samples is enforced before a new event can fire. Recovery is signaled when the rolling accuracy returns to the pre-drift baseline (estimated over the window preceding the drift) or after a timeout of 2Δ samples, after which τ is annealed back toward its steady-state value at rate $\Delta\tau$ per step and η decays linearly to η_0 over Δ steps.

The two-stage design balances a confirmation-only reaction (too slow under sudden drift) against pre-warming without a gate (amplifies noise). The reset in (iii) matters for the adaptive constraint: the price of Section IV-B tracks the *current* disparity, but the regularizer's direction comes from statistics accumulated since the last reset, so without forgetting a correctly raised price would push along the pre-drift direction. For the same reason a *fairness monitor*: one ADWIN detector per group on the per-sample signal $v_t[a]$ of (10) also resets the statistics when a group's disparity shifts; it does not change the learning rate or the temperature. Because the signal is unbiased and not autocorrelated, this detector avoids the false alarms that a rolling-window disparity would cause. Concrete values for all controller parameters are reported in Table II.

D. Datasets and Preprocessing

Table I summarizes the two datasets. FOLKTABLES (ACS Income, CA) [27] supplies a large natural cross-year shift between 2015 (training) and 2017–2018 (prequential test); ADULT [28] provides a controlled testbed on which we inject concept-drift perturbations of a known form on the test stream. Both datasets are processed through a shared tabular pipeline: numerical features are standardized using training-set means and standard deviations, and categorical variables are one-hot encoded. The encoder is fitted on the training split alone and reused unchanged for the test stream, so the streaming protocol remains free of test-time information leakage.

E. Concept-Drift Scenarios

We adopt complementary strategies on the two datasets. On FOLKTABLES, training on 2015 and evaluating prequentially on 2017–2018 already provides a real, multidimensional cross-year shift driven by genuine demographic and economic change; injecting additional synthetic perturbations on top of it would mask controlled and natural signals obscuring the source of any observed degradation.

TABLE I
DATASETS

Dataset	Size	Target	Protected attribute	Split / stream
FOLKTABLES (ACS Income, CA)	187,474 + 187,475	Income $>$ \$50k	Sex (2 groups)	Train 2015; test 2017–2018 in temporal order; both splits uniformly subsampled to 10% per seed
ADULT	32,561 + 16,281	Income $>$ \$50k	Sex (2 groups)	Standard train/test split; one injected drift per test stream

On ADULT the test stream carries no known drift, so we inject it. We use four drift types [12], each giving a separate scenario: every model ran on all four, with one drift per stream at an onset drawn uniformly between 50% and 70% of the stream. The four types are *x_permutations*, where the feature vector is permuted by a fixed random permutation, changing the mapping from inputs to labels; *y_swaps*, where samples of one randomly chosen class are relabeled with probability 0.5, a concept drift in $P(y \mid x)$; *y_prior_skip*, where 75% of the samples of one class are dropped, a prior shift; and *x_exceed_skip*, where samples whose value of a randomly chosen numerical feature exceeds its training median are dropped, a covariate shift. The onset, the drifted class, the permutation, and the feature are drawn from the seed, so each seed is a different stream; the order of the ADULT test split is fixed.

F. Baselines

We compare FADO against three streaming baselines that span the accuracy/fairness/adaptivity design space:

- **ARF** [9]: the Adaptive Random Forest from the River library [29], used as an accuracy-oriented streaming reference that handles drift through per-tree warning and drift detectors and background trees, but without any fairness mechanism;
- **RFR**: a sharpness-aware reweighting variant that applies the RFR optimizer [13] (sharpness-aware minimization over a fairness-reweighted loss); this provides a fairness-aware static baseline without adaptive drift handling;
- **ARANYANI** [1]: the plain ARANYANI forest.

All baselines are evaluated under the same $batchsize = 1$ test-then-train prequential protocol as FADO so that comparisons are not biased by data ordering or data leakage. The global random number generators (NumPy, Python `random`, and TensorFlow) are pinned per seed and TensorFlow op determinism is enabled, so the data split, the forest initialization, and the training trajectory are identical across methods that share a seed.

G. Hyperparameter Selection

Table II lists the fairness and controller hyperparameters with their ADULT defaults and the FOLKTABLES overrides.

For the base learner we keep the settings of [1]: $M = 3$ trees of depth $h = 4$, trained with Adam at $\eta_0 = 2{\times}10^{-3}$. The remaining values were set before the reported runs and were never tuned on the reporting seeds.

As aforementioned, the fairness target $\varepsilon = 0.05$ is a fixed requirement rather than a tuned value: for two groups it allows a positive-prediction-rate gap of at most 0.10. The dual step $\eta_\lambda = 0.01$ trades responsiveness against noise in the per-sample signal (Remark 1), and the cap $\lambda_{\max} = 10$ bounds the accuracy that the price can spend when ε is hard to reach (again, the upper bound is not required and we show it's never actually reached). The surrogate transition is $\delta = 0.1$, larger than the 0.01 used in [1].

The two ADWIN levels make the warning detector fire first: in ADWIN a larger δ is more sensitive, so $\delta_w = 2{\times}10^{-2}$ gives an early warning, while $\delta_c = 10^{-5}$ requires a strong, sustained rise of the error rate before the full reaction starts. The learning-rate multipliers β_w=5 and β_c=10 keep a 1:2 ratio: a warning gives a measurable pre-warm without committing the model to a new regime, and only a confirmed drift triggers the full spike. The drift temperature $\tau_{\text{drift}} = 0.1$, ten times lower than the steady-state $\tau = 1$, makes routing nearly hard during the spike, so the forest can move to a new leaf assignment within a few dozen samples instead of waiting for the soft routing weights to average over the new regime. The anneal $\Delta\tau = 2{\times}10^{-3}$ returns τ from 0.1 to 1 in 450 steps, well inside the $\Delta = 3000$ learning-rate decay horizon. The cooldown $C = 200$ stops the controller from firing again during its own response, and the label-noise guard $\nu = 0.7$ keeps wrong predictions from triggering an event. Because FOLKTABLES's drift is a gradual change between survey years rather than an abrupt break, we fixed a milder reaction for it in advance: a smaller spike ($\beta_c = 3$), a softer temperature ($\tau_{\text{drift}} = 0.5$), and a shorter horizon ($\Delta = 1000$), which avoid overreacting to a slow change.

TABLE II
FADO HYPERPARAMETERS

Symbol	ADULT	FOLKTABLES[a]	Role
Fairness regularizer and constraint			
λ_{base}	0.1	–	floor of λ_t; ARANYANI's fixed λ
ε	0.05	–	demographic-parity target
η_λ	0.01	–	dual step size
$\lambda_{\max}$	10	–	cap on λ_t
δ	0.1	–	quadratic/linear transition in ϕ_δ
W	1000	–	rolling window for reported metrics
Drift detection and reaction controller			
δ_w	2×10^{-2}	–	ADWIN significance (pre-warm)
δ_c	1×10^{-5}	–	ADWIN significance (confirmation)
δ_{fair}	2×10^{-3}	–	ADWIN significance (fairness monitor)
β_w	5	–	learning-rate multiplier on warning
β_c	10	3	learning-rate multiplier on confirmed drift
Δ	3000	1000	learning-rate decay horizon back to η_0
τ	1.0	–	steady-state routing temperature (added by FADO)
τ_{drift}	0.1	0.5	temperature set on confirmed drift
$\Delta\tau$	2×10^{-3}	–	per-step temperature anneal back to τ
C	200	–	cooldown (samples) between events
ν	0.7	–	label-noise confidence guard
m_0	30	–	minimum samples before first activation

[a] A dash means the FOLKTABLES value equals the ADULT value.

H. Evaluation Protocol

All methods are evaluated under a strict prequential, $batchsize = 1$, test-then-train protocol [30]: at each step the model emits $\hat{y}_t$, the prequential metrics are updated, and only then is the parameter update performed using the just-observed (y_t, a_t). We report mean and standard deviation over 30 reporting seeds, disjoint from the seeds used during development. The primary metrics are prequential accuracy and the demographic-parity violation, with equalized odds as a secondary, unoptimized criterion, all computed over the same W-sample rolling window and averaged over the whole stream.

For each metric column, paired-by-seed comparisons of FADO against each baseline are performed using the two-sided Wilcoxon signed-rank test; p-values are corrected within each column using the Holm–Bonferroni step-down procedure across the three baselines. The procedure follows the recommendations of Demšar [31]. We prefer the paired Wilcoxon test to the Welch two-sample t-test because it assumes no distribution for the per-seed metric and uses the pairing given by the shared data ordering (Section IV-F), which Welch ignores. The Welch results are reported alongside in the repository (Section IV-I).

I. Reproducibility

The full implementation, configuration files, seed manifests, and per-seed result files are released at https://github.com/TiagoSilva35/JustFADO. The repository's README reproduces Table III and lists, for each dataset, the exact command-line invocation that produced every reported row (per-seed sweep, paired Wilcoxon and Welch significance run, summary aggregation). Running the commands as listed regenerates the same per-seed result files that the reported means, standard deviations, and Holm-corrected p-values are computed from.

V. RESULTS AND DISCUSSION

We report the prequential results on the natural cross-year shift of FOLKTABLES and on the four injected drifts of ADULT (Table III), as mean $\pm$ standard deviation over 30 seeds, with the demographic-parity regularizer as the optimization target. All metrics are averaged over the whole stream.

TABLE III
PREQUENTIAL ACCURACY, DEMOGRAPHIC PARITY, AND EQUALIZED ODDS OVER 30 SEEDS[a]

Method	Acc. ↑	DP ↓	EO ↓
FOLKTABLES, natural cross-year shift			
FADO	0.805 ± 0.003	$\mathbf{0.042 \pm 0.003}^{***}$	$\mathbf{0.025 \pm 0.003}^{***}$
ARANYANI	$\mathbf{0.808 \pm 0.003}$	0.057 ± 0.004	0.036 ± 0.003
ARF	0.731 ± 0.004	0.072 ± 0.012	0.077 ± 0.016
RFR	0.779 ± 0.003	0.075 ± 0.006	0.059 ± 0.006
ADULT, x_permutations			
FADO	$\mathbf{0.838 \pm 0.003}$	$\mathbf{0.061 \pm 0.004}^{***}$	$\mathbf{0.053 \pm 0.009}^{***}$
ARANYANI	0.837 ± 0.004	0.083 ± 0.007	0.073 ± 0.010
ARF	0.826 ± 0.004	0.090 ± 0.003	0.110 ± 0.013
RFR	0.837 ± 0.002	0.100 ± 0.004	0.091 ± 0.010
ADULT, y_swaps			
FADO	**TBD**	**TBD**	**TBD**
ARANYANI	**TBD**	**TBD**	**TBD**
ARF	**TBD**	**TBD**	**TBD**
RFR	**TBD**	**TBD**	**TBD**
ADULT, y_prior_skip			
FADO	**TBD**	**TBD**	**TBD**
ARANYANI	**TBD**	**TBD**	**TBD**
ARF	**TBD**	**TBD**	**TBD**
RFR	**TBD**	**TBD**	**TBD**
ADULT, x_exceed_skip			
FADO	**TBD**	**TBD**	**TBD**
ARANYANI	**TBD**	**TBD**	**TBD**
ARF	**TBD**	**TBD**	**TBD**
RFR	**TBD**	**TBD**	**TBD**
ADULT, mean of the four drifts			
FADO	**TBD**	**TBD**	**TBD**
ARANYANI	**TBD**	**TBD**	**TBD**
ARF	**TBD**	**TBD**	**TBD**
RFR	**TBD**	**TBD**	**TBD**

[a]Mean ± standard deviation over 30 seeds, over the whole stream. Bold marks the best mean per metric within each block. Superscripts on FADO mark a paired Wilcoxon test, Holm-corrected, significant against *every* baseline in FADO's favor ($^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$); columns with no marker have mixed per-baseline directions and are dissected in the text. **TBD** marks runs still in progress.

A. To What Extent Does Concept Drift Affect Group Fairness?

B. Can Adaptive Mechanisms Preserve Fairness While Maintaining Utility?

C. Ablation of the Controller Components

VI. CONCLUSION AND FUTURE WORK

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